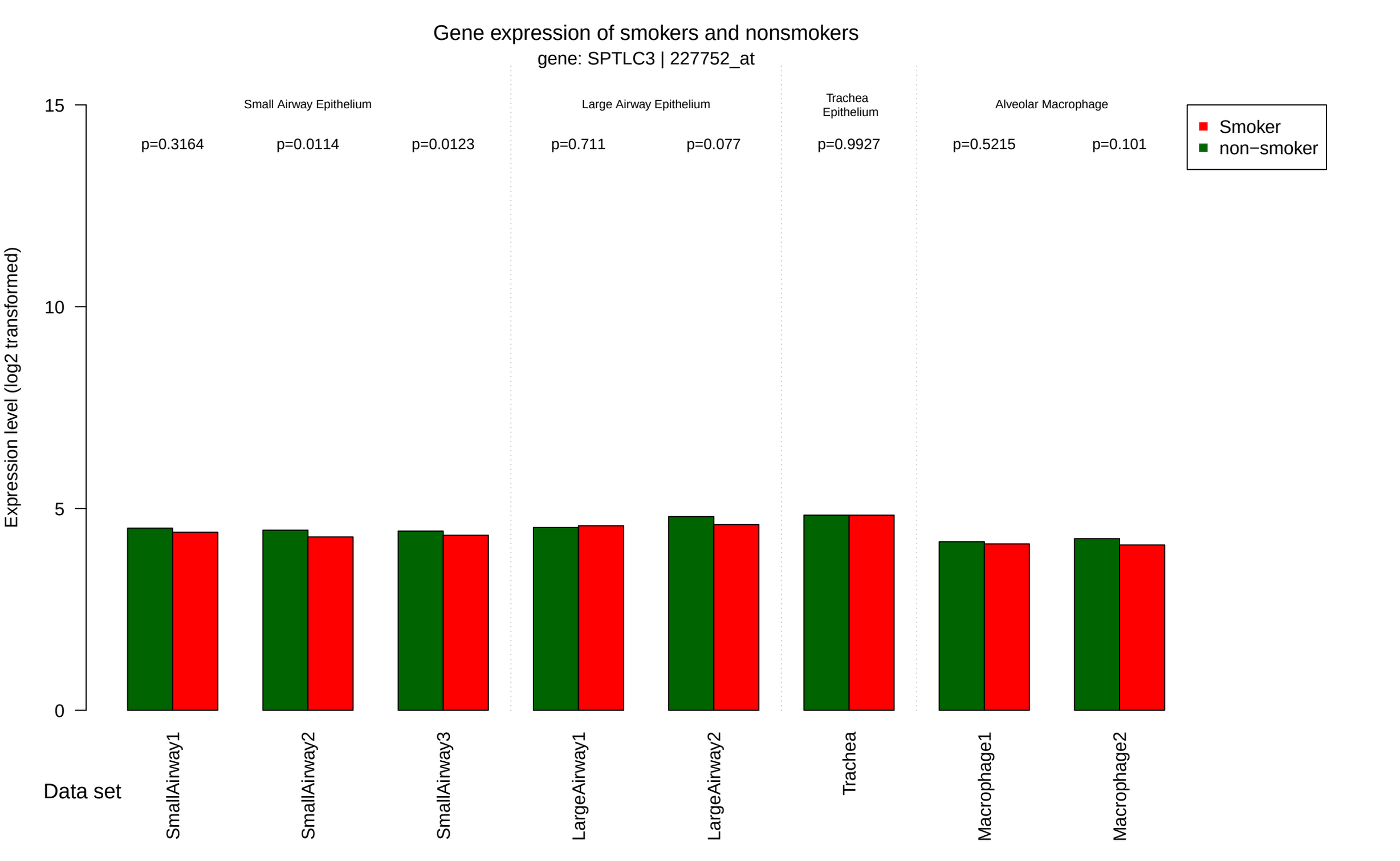
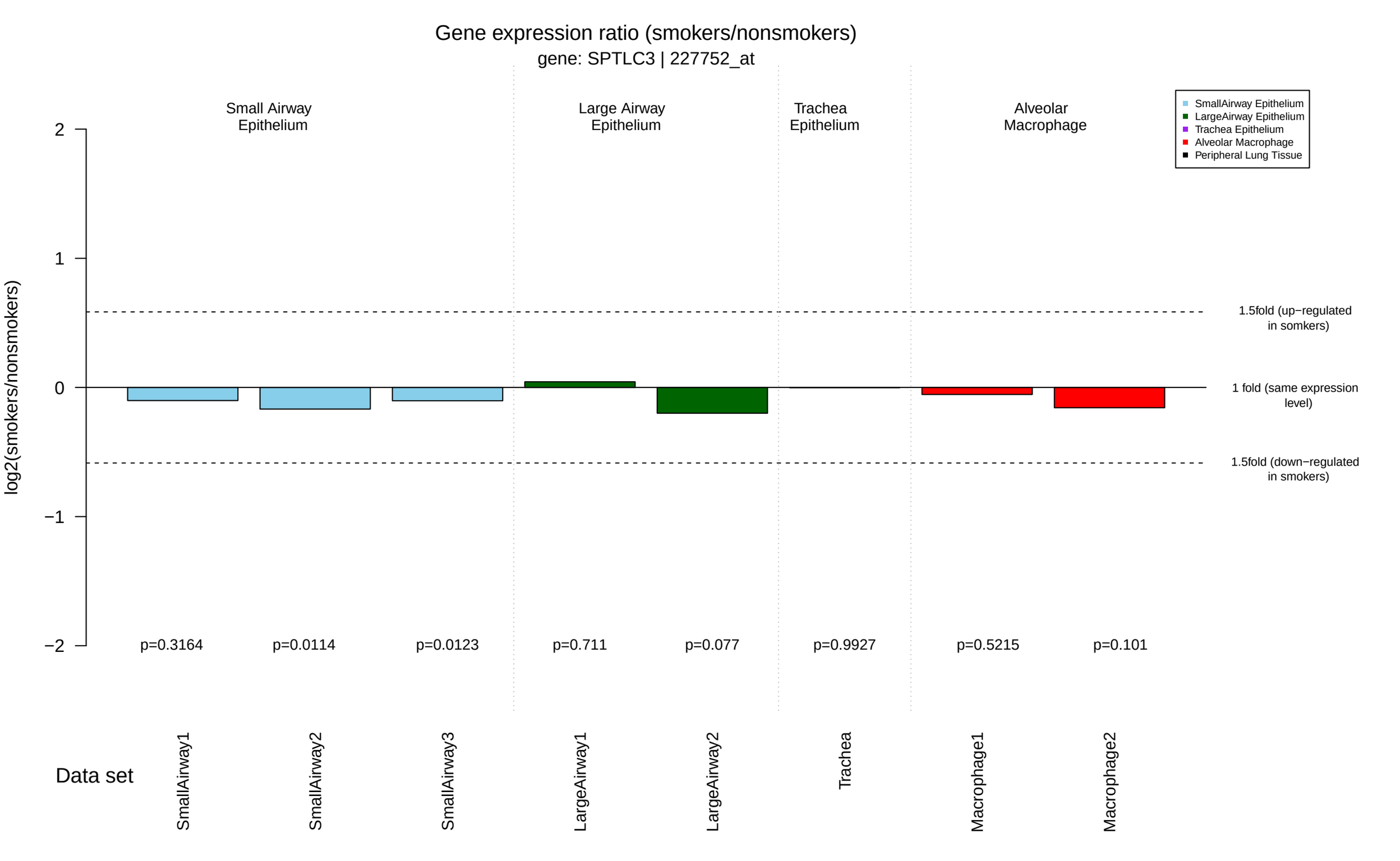
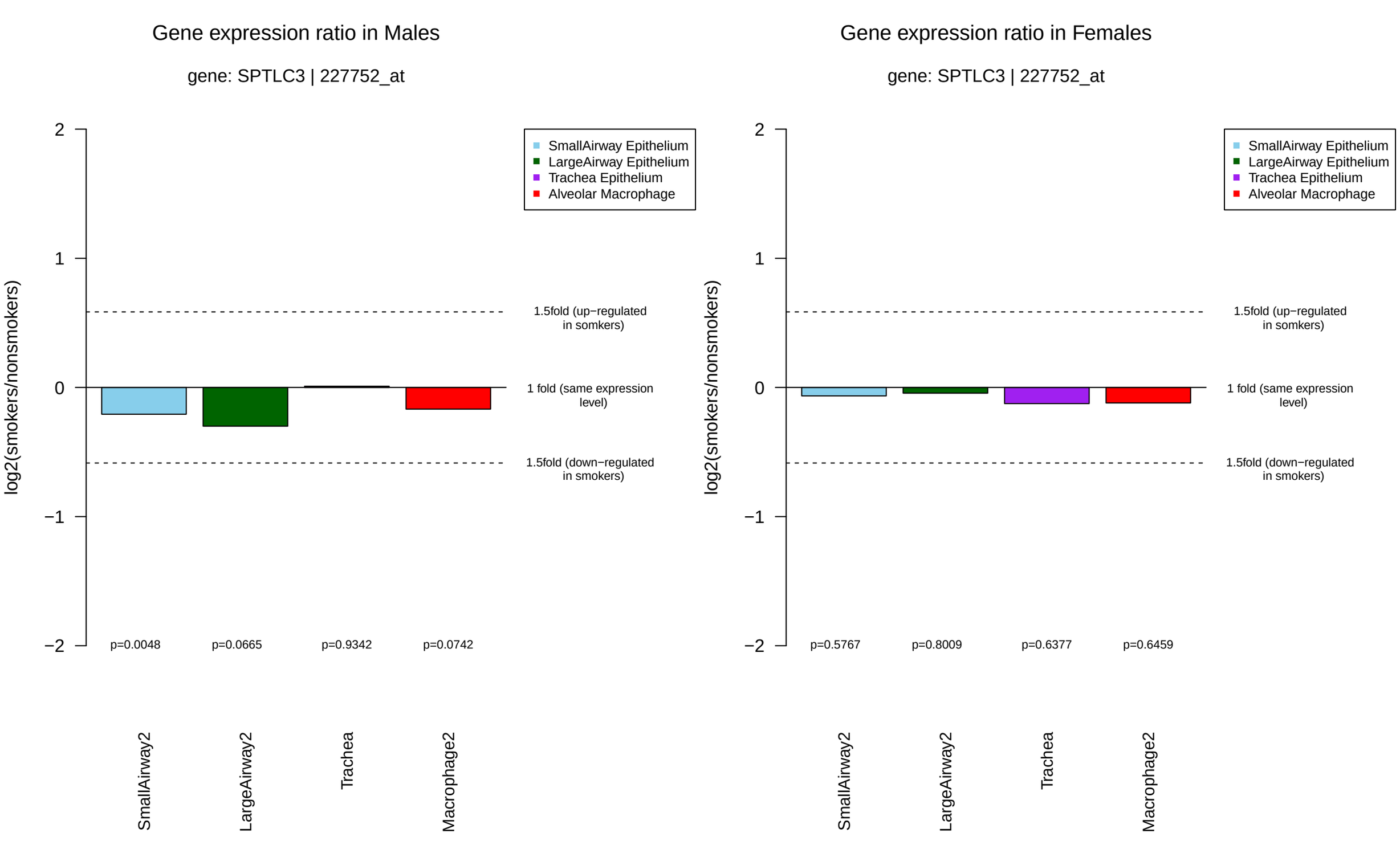
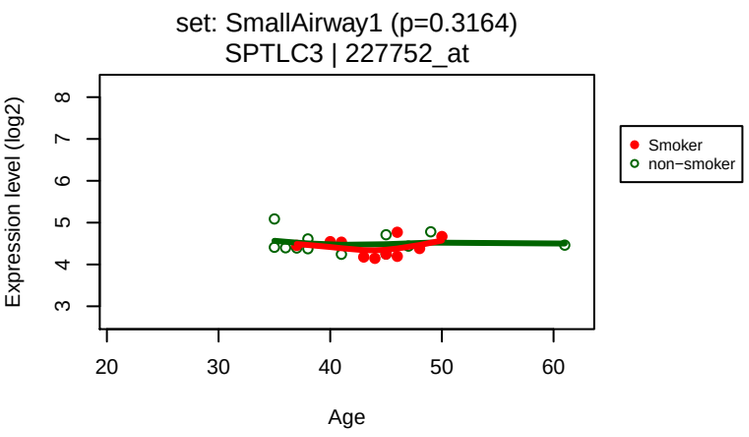
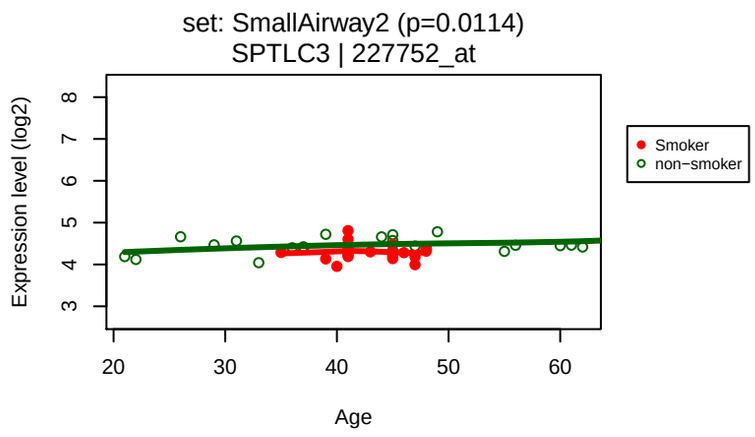
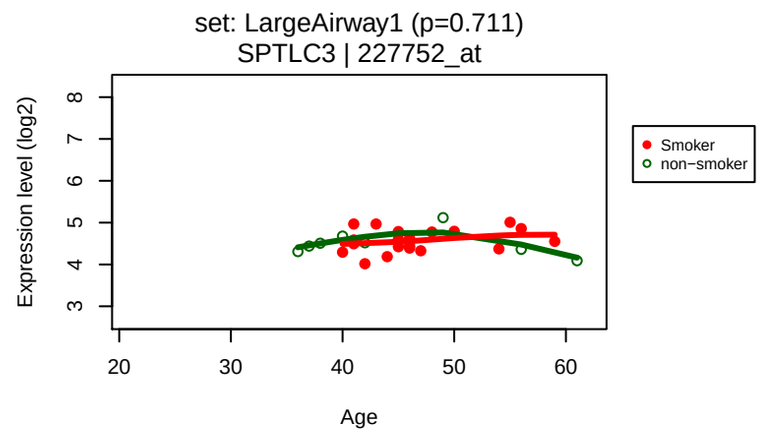

Gene expression of smokers and nonsmokers
gene: SPTLC3 | 227752_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3164
p=0.0114
p=0.0123
p=0.711
p=0.077
p=0.9927
p=0.5215
p=0.101
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

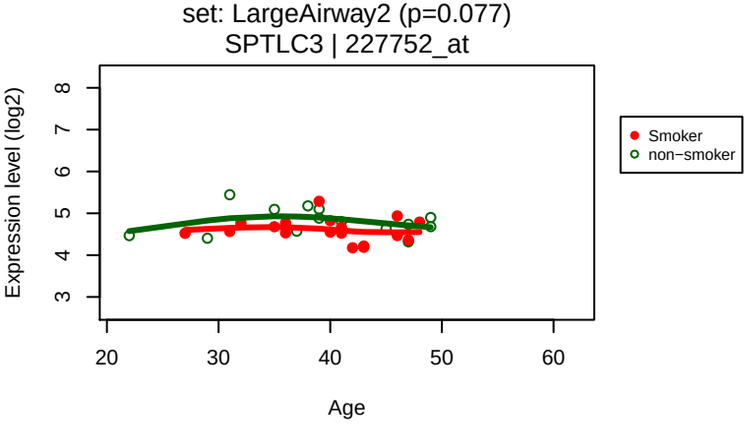

Gene expression ratio (smokers/nonsmokers)
gene: SPTLC3 | 227752_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3164
p=0.0114
p=0.0123
p=0.711
p=0.077
p=0.9927
p=0.5215
p=0.101
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

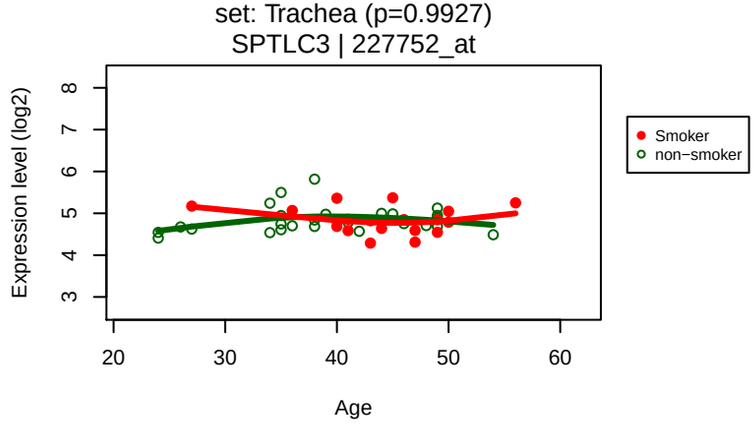

Gene expression ratio in Males
gene: SPTLC3 | 227752_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0048
p=0.0665
p=0.9342
p=0.0742
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPTLC3 | 227752_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5767
p=0.8009
p=0.6377
p=0.6459
SmallAirway2
LargeAirway2
Trachea
Macrophage2

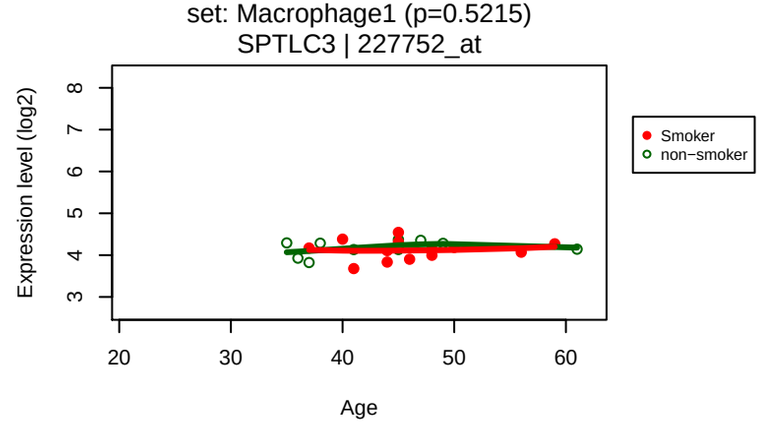

set: SmallAirway1 (p=0.3164)
SPTLC3 | 227752_at
Expression level (log2)
Age
Smoker
non−smoker

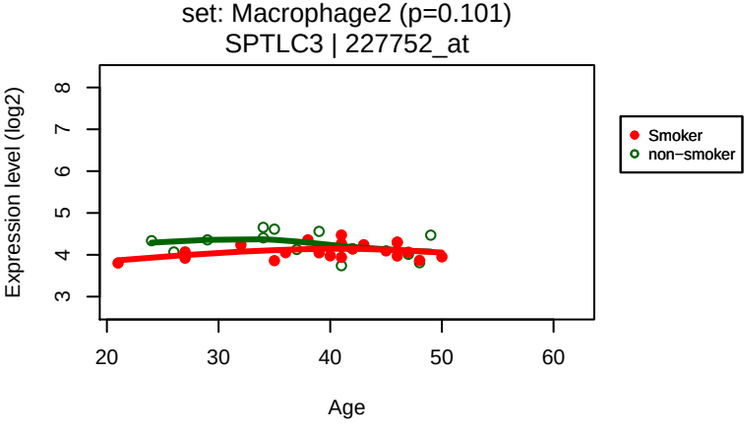

set: SmallAirway2 (p=0.0114)
SPTLC3 | 227752_at
Expression level (log2)
Age
Smoker
non−smoker

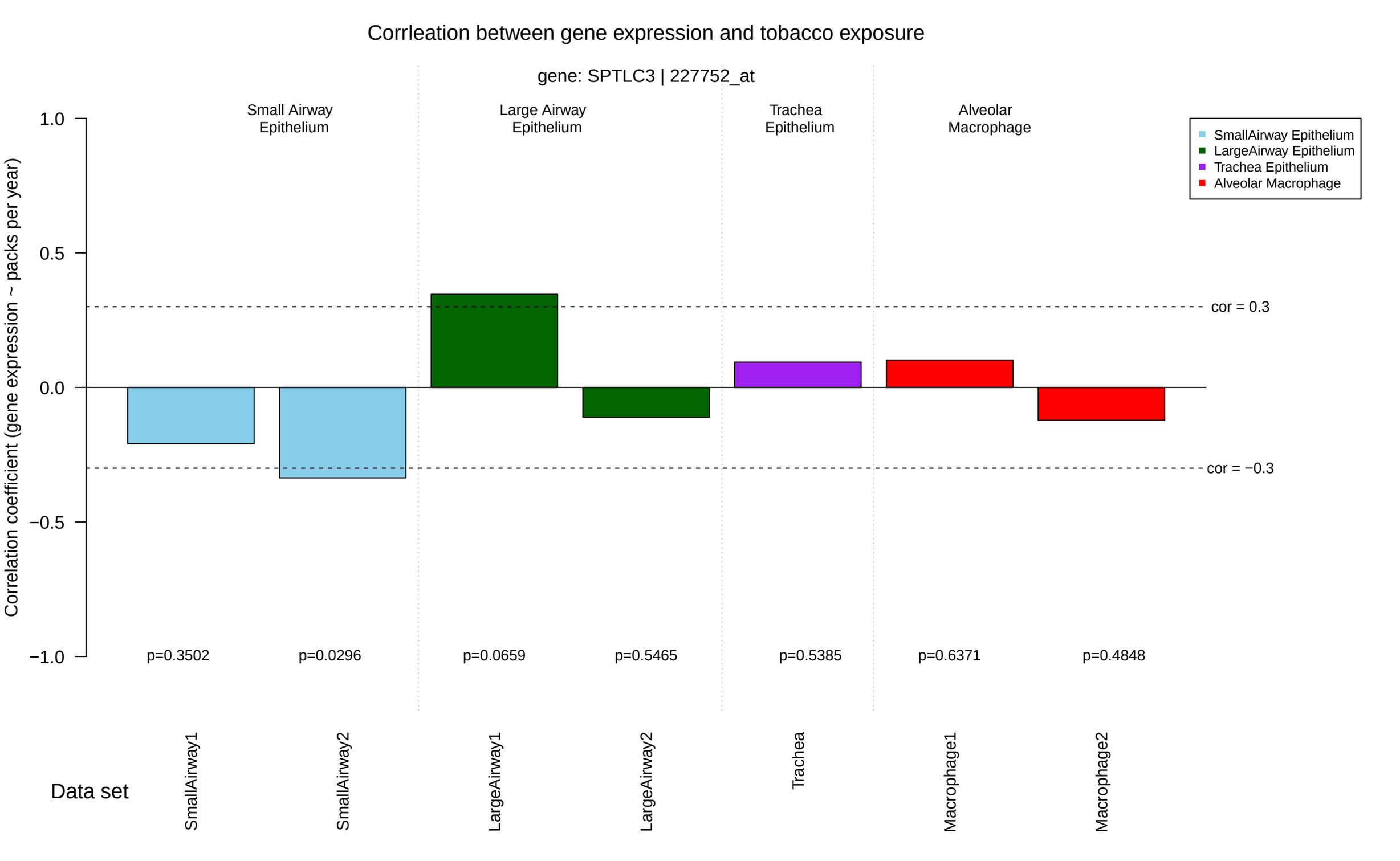

set: LargeAirway1 (p=0.711)
SPTLC3 | 227752_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.077)
SPTLC3 | 227752_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9927)
SPTLC3 | 227752_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5215)
SPTLC3 | 227752_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.101)
SPTLC3 | 227752_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPTLC3 | 227752_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3502
p=0.0296
p=0.0659
p=0.5465
p=0.5385
p=0.6371
p=0.4848
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2